

Replication Report: (title not recorded)

OSTI 1484740 | Coverage 8/10, Agreement 8/10

Rick Stevens

Ollie (AI Assistant)
Argonne National Laboratory

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1 Source paper

- **Title:** (title not recorded)
- **Authors:** Not recorded
- **Year:** n/a
- **Domain:** Not recorded
- **Identifier:** OSTI 1484740

2 Replication objective

The central claim of the paper concerns not recorded. Our objective was to reproduce the headline numerical/qualitative results within the constraints of the AI-driven Replication Project (24–72 h, commodity GPU/CPU compute, no author contact).

3 Methodology

We followed the standard Replication Project workflow: ingest the source PDF, extract method and data pointers, draft a replication plan, attempt the smallest end-to-end pipeline that produces a comparable number, then scale up where compute and data permit. Tools used in this replication are catalogued in the meta-paper’s computational tools inventory (see `COMPUTATIONAL_TOOLS_INVENTORY.md`).

This paper went through the Tier-Lift pipeline; supplementary notes at `.openclaw/workspace/24h-progress/bat`, `.openclaw/workspace/24h-progress/tier-lift-v2.5/1484740/REPORT.md`, `.openclaw/workspace/24h-progress`.

The replication was driven by an OpenClaw agent loop (Claude Opus / GPT-5 class models) issuing shell, Python, and (where applicable) GPU jobs. Compute usage and wall-time for individual papers are summarised in the meta-paper’s resource table; not separately recorded for this paper unless noted in the Tier-Lift artefact. Sub-agents were spawned for replication-plan generation and (for papers in batches 1–4) for tier-lift extension runs.

4 What was reproduced

- Not recorded.

5 What was missing or incomplete

- Not recorded.

The dominant blocker categories for this paper, mapped onto the meta-paper’s 9-category friction taxonomy (§4), are listed in §?? below.

6 Quantitative agreement

Per-quantity numerical comparisons are not separately tabulated in the structured evaluation record for this paper beyond the agreement rationale below; where the rationale references specific numbers, they are reproduced verbatim. A full numerical comparison table is recorded in the per-replication artefacts (see §??). For papers below 8/8, the agreement rationale identifies the specific quantities that diverge.

Agreement rationale. Not recorded.

7 Score and friction-point tagging

- **Coverage:** 8/10 — Not recorded.
- **Agreement:** 8/10 — Not recorded.

Friction points encountered (taxonomy tags):

- None recorded.

8 Follow-on questions

- Not recorded.

9 Reproducibility pointers

- **Paper directory:** 1484740-Electronic-and-optical-properties-of-two-dimensional-GaN
- **Replication artefacts:**
 - /Users/stevens/Dropbox/REPLICATE-PROJECT/1484740-Electronic-and-optical-properties-of-
- **Replication-dir hint from JSONL:** /Users/stevens/Dropbox/REPLICATE-PROJECT/1484740-Electronic-
- **Status:** tier_lift_v2.5_2026_04_27

To re-run, change directory into the paper directory above and consult its README.md for the entry-point script. Most replications follow the convention `replication/run_*.sh` or `replication/<subdir>/run.py`.